Greenfoot Project “Shell Out!”:
Project Prospectus

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Game Description

Shell Out! is an arcade-style obstacle-avoidance game inspired by Frogger. Players control a small cartoon-style turtle attempting to navigate across a dangerous road to reach the safety of its home. Along the way, the turtle must avoid moving vehicles and carefully time its movements to survive. In addition to reaching home safely, the turtle must rescue its three missing children, represented by smaller green turtles who are scattered across the road, before a hungry flock of ravenous birds catches up to them. Players must successfully collect all three baby turtles and make it home safely within sixty seconds to win the game. Players earn points for rescuing their children and successfully reaching home, with additional points to be gained for speed of completion. Collisions with vehicles result in lost lives, and running out of lives before completing the rescue results in a game over. A sixty-second countdown timer adds another challenge, and running out of time before completing the rescue also results in a game over as the birds steal the remaining baby turtles.

Player Turtle

The Player Turtle is the main character and is controlled by the player. It will be represented by a cartoon-style green turtle with a green shell (turtle2.png). The player uses the keyboard W, A, S, and D keys to move the turtle up, down, left, and right across the game world, respectively. Its primary objective is to avoid traffic, rescue its three children, and safely reach its home before time runs out. If the player collides with a vehicle, a life is lost, and the turtle returns to its starting position.

Baby Turtles

The Baby Turtles are three smaller versions of the main character, each represented by a small green turtle sprite (turtle.png). They will be positioned at random locations throughout the road and must be collected by the player. When the Player Turtle makes contact with a Baby Turtle, that child is rescued and thus disappears from the playing field, and the rescue counter increases by one. All three children must be collected before the player can successfully complete the game.

Vehicles

Vehicles serve as the primary obstacles throughout the game. They will be represented by cartoon-style cars and trucks (randomly selected sprites from the transport library of Greenfoot's image repository) that move horizontally across the screen in designated lanes. Vehicles will travel at different speeds and directions, requiring the player to carefully time their movements. If the Player Turtle collides with a vehicle, the player loses a life and must attempt the crossing again.

Ravenous Birds

The Ravenous Birds are the antagonists of the game and threaten the safety of the three Baby Turtles. They will be represented by cartoon-style pelicans with large beaks. The birds serve as a visual explanation for the sixty-second countdown timer. If the player fails to rescue all three Baby Turtles before the time expires, the birds enter the game world and land on the Baby Turtles, stealing the remaining children and resulting in a game over.

Home

The Home location represents the Player Turtle's destination and the safe place where the family can reunite. It will be represented by a small grassy area at the other side of the road. Once the player has collected all three Baby Turtles, they must successfully navigate to the Home location and reach it before the countdown reaches zero. Reaching home with all three children completes the game and displays a victory screen containing the player's final score and game statistics.

References/Links

https://github.com/yataraou/-CSC170-Greenfoot-Project-ShellOut

[CSC170] Greenfoot Project: Game Instruction Document